

RAGLoom

A visual RAG pipeline editor for local, on-device product Q&A.

Drop in product spec documents, ask questions in natural language, and get grounded answers — generated entirely on-device, with no cloud dependency. Built in Python, FastAPI, React and ChromaDB on Ollama. **No LangChain.**

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Source-available · Commercial licensing: tsoleou@gmail.com · Product Brief · v2

THE PROBLEM

How do you run a trustworthy product-Q&A demo on a laptop, with no internet and a small model — without it inventing specs or recommending a competitor?

Trade-show booths and retail kiosks want a live, conversational product assistant — and for a PC OEM, running the model on the device itself doubles as proof of AI-PC performance. But the local-only constraint forces a small model (`gemma3:4b`), and small models hallucinate, misread numbers, and can't be trusted to "just follow instructions" on the things that matter at a booth: never quoting a price, never naming a competitor, never recommending a 1.8 kg laptop for a "under 1 kg" request.

THE APPROACH

RAGLoom moves every high-stakes decision **out of the prompt and into code**. Four deterministic gates short-circuit a query *before* it can reach the generator, each firing on a different signal and composing without overlap. What's left for the LLM is the one thing it's good at: phrasing a grounded answer from retrieved context.

The whole pipeline is a **visual, drag-and-drop node graph** — so a PM can open the editor mid-demo, point at the safety layer, and watch a blocked query light up amber across the canvas in real time.

THE SAFETY LAYER — POLICY IN CODE, NOT IN THE PROMPT

● Guardrail

Keyword block list (e.g. competitor brands) with word-boundary matching → canned refusal.

● PriceGuard

Regex price-intent detector (price / MSRP / 多少錢...) → bilingual refusal, pre-retrieval.

● ScopeGate

Semantic on/off-topic check via anchor embeddings. Survives bridge attacks pure cosine can't.

● ConstraintFilter

Numeric spec gate ("under 1kg"). Resolves each candidate's canonical spec, drops violators.

At small-model scale, "trust the model to follow instructions" is unreliable for refusals and numeric comparison. A 4B model will mark `1.8 kg < 1 kg` as passing — so the comparison never reaches it. Each gate degrades safely and, when it fires, returns a canned bilingual refusal rather than handing the model an empty context to hallucinate over.

THE PRODUCT

WHO IT'S FOR

End users / booth visitors

A Chat UI: ask a question, get a formatted, grounded answer with a retrieval-details panel and (in chatbot mode) an animated avatar that reflects the model's self-reported emotion. Replies always come back in the visitor's language.

Builders & operators

A Node Editor to assemble, rewire and A/B the pipeline, plus an in-editor batch eval harness to measure retrieval quality before shipping a graph variant.

CONVERSATIONAL INQUIRY FLOW — A BOOTH ASSISTANT, NOT A SEARCH BOX

`IntentRouter` classifies each visitor turn (one small LLM call) into one of four inquiry intents, re-detected *every turn* so a topic-hopping visitor is followed immediately. `DialogueFlow` then runs the matching multi-stage script, advancing the conversation toward a recommendation:

SPEC	Answer the asked spec from the KB
RECOMMEND	Elicit needs → Recommend 1–2 models with a reason
COMPARE	Confirm which models → List key differences + a fit tip
SUITABILITY	Clarify the use case → Judge fit; suggest an alternative if not

Guard-aware state. Stage and intent only advance on a *committed* turn — one where the generator actually produced an answer. A guard short-circuit freezes the whole funnel, so refusals and off-topic turns never pollute the conversation's stage or history. The single advance gate (`decide_advance`) is the one place designed to swap LLM judgement for deterministic slot logic — hardening a script without touching anything else.

CAPABILITIES — WHAT'S IN THE BOX

● Visual node editor v2

27 node types across ingest / query / eval.
Real-time per-node execution status over WebSocket; first-class edge editing. **LLM-invoking nodes carry a distinct background tint** — the compute-heavy steps are obvious at a glance.

● Dialogue-aware routing

IntentRouter + DialogueFlow turn the pipeline into a guided booth conversation — four inquiry scripts with guard-frozen state, shipped as the active `booth_inquiry` profile.

● Metadata-filtered retrieval

Docs tagged with a `product_id` at ingest. A ProductSelector routes point-queries to a single product (rule mode = zero LLM latency); comparisons fall through to broad search.

● Retrieval Judge

An LLM rerank pass drops retrieved chunks that don't actually answer the query — catching polarity / negation misses pure cosine retrieval can't. Degrades to keep-all on error.

● Always-on reference data

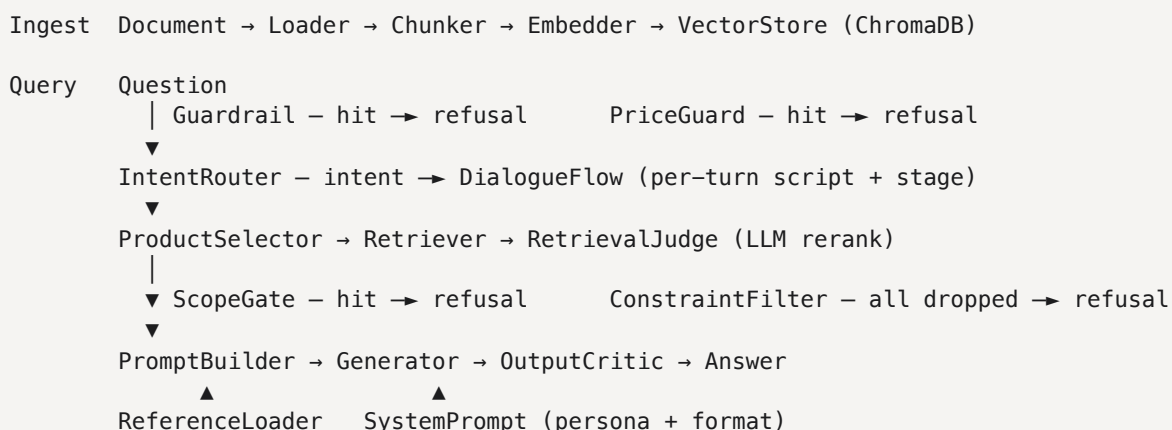
A static comparison table injected into every prompt, guaranteeing broad coverage for comparison queries independent of vector retrieval.

● Persona presets & structured output

professional / chatbot registers tuned to a trade-show promoter voice. Chatbot mode emits grammar-constrained JSON `{reply, emotion}`, rendered as a themeable avatar.

UNDER THE HOOD

ARCHITECTURE



QUALITY & EVALUATION

- **Golden-set eval with LLM-as-judge** — a curated regression set, audited by a second LLM that returns explicit `hallucinated_claims` lists. Gates on the binary signal (claims empty / non-empty),

not noisy float scores, so same-commit reruns don't flicker pass/fail.

- **In-editor batch eval** — a dedicated eval-node family (Hit@K coverage, score distribution, diversity, facts coverage) plus a Run Batch sweep over golden-set cases, with macro averages, per-category breakdown and worst-K ranking — fast, deterministic, retrieval-only.
- **One-click report export** v2 — the batch panel renders a self-contained, colour-coded **HTML report** (or Markdown) straight from the on-screen results, matching the `eval/report.py` CLI output — no re-run of the sweep.
- **Operator-readable metrics** v2 — score-distribution surfaces as a colour-coded **檢索信心 (retrieval confidence)** reading (top-1 similarity), with hover hints on every metric, so a non-technical booth operator can read retrieval health at a glance instead of a blank cell.
- **Hardened local API** — every endpoint requires an auto-generated local token, CORS is locked to the local origin, file-path params are confined to an allowlist, and batch eval is bounded. A stray malicious browser tab can't drive your local pipeline.

TECH STACK

Python · FastAPI · React · Vite · TypeScript · ChromaDB · Ollama (gemma3:4b) · nomic-embed-text · WebSocket · No LangChain

AT A GLANCE

